Электроника МКШ-2 Operating Manual — English Translation

Nov 5, 2026 · @Brett

Translator's note

This is an English translation of the operating manual (Руководство по эксплуатации, abbreviated РЭ) supplied with the Электроника МКШ-2 school microcalculator. It was made by Kalkulyator in Svetlovodsk, Kirovograd Oblast, Ukrainian SSR. The original runs to 32 pages; the section and table numbering below follows it.

- **Key names** are given as printed on the keyboard, for example **Сх** (clear) and **ВП** (enter exponent), with their meaning on first use.
- **Units and number formats** are kept as in the original, but the decimal comma is shown as a decimal point.
- **Translator's additions** are marked [in square brackets].
- **Figures:** the drawings of the calculator's exterior and the mounting template are described in words. The circuit diagrams in Appendix 1 are summarised, with all labels and notes translated.

1. Attention!

1.1. Before starting work with the microcalculator, read this operating manual (РЭ) carefully.

1.2. A special feature of the microcalculator is that it runs from an **alternating-current supply of 42 V**.

1.3. The tear-off coupons for warranty repair are cut out by the repair organisation's staff only after the repair has been completed.

1.4. The manufacturer's staff work continually to improve the technical and operating characteristics of the microcalculator, and would be grateful for your feedback on its performance (Appendix 2). Please send the completed feedback sheet to:

317000, Svetlovodsk, Kirovograd Oblast, P.O. Box No. 7.

2. Brief description

2.1. The «Электроника МКШ 2» microcalculator is intended for use in the classrooms of general secondary schools and other educational institutions, for studying computing equipment and for solving problems from the secondary-school curriculum.

2.2. The microcalculator performs the following operations:

- changing the form in which a number is displayed;
- changing the sign of a number or of its exponent;
- the four arithmetic operations;
- calculating the functions x², 1/x, √x, y^x, e^x, 10^x, ln x, lg x, sin x, cos x, tg x, arcsin x, arccos x and arctg x, where the argument of the trigonometric functions can be in degrees or radians depending on the position of the Г–Р (degrees–radians) switch;
- storing information in the memory register;
- recalling information from the memory register;
- accumulating data or results in the memory register;
- exchanging the contents of the memory register and the display register;
- clearing the display register;
- cancelling the second-function (shift) flag;
- calculating the roots of a quadratic equation;
- calculating the roots of a system of two linear equations in two unknowns;
- entering an exponent;
- calculations with a constant.

2.3. Overflow, and the result of an invalid operation, are shown on the display as the sign **«—» in all digit positions**. Invalid operations occur when a function is calculated with an argument outside its permitted range (Table 1).

2.4. The arrangement of a number on the display is shown in Figure 1.

2.5. When any key except the auxiliary **F** key is pressed, the symbol «.» may light in the service digit position. [This is normal.]

2.6. Precious metals contained in the microcalculator: gold 0.03753 g; silver 0.16456 g.

2.7. The non-ferrous metal content of the microcalculator is given in Table 1a.

Table 1a. Non-ferrous metal content

Part or assembly	Metal or alloy	Grade	Mass, g	Note
1. Bushings (in the cover)	Brass	ЛС59-1	9.2	4 pcs
2. Display	Tungsten	ВА	0.135	
	Tungsten alloy	МВ-50	0.022	
	Nickel	НП2	2.652	
	Precision alloy	42Н	0.060	
	Platinite	ПТБ-2	1.3	
3. Jumpers (between boards)	Copper	ММО,6	1.29	From the main (операционное) board to the input (keyboard) board
4. Jumpers (between boards)	Copper	НВ-0,12	0.349	From the power supply to the main board
5. Pin	Brass	Л63	2.1	Mains-lead plug

Note: 1 cm of display external lead contains 0.0018 g of copper and 0.0028 g of nickel.

Figure 1. Arrangement of a number on the display

The display has nine positions, numbered 1 to 9 from left to right.

- **(a) Normal (exponential) form:** position 1 is the sign of the number, then the mantissa, then the sign of the exponent and the two-digit exponent in the last positions.
- **(b) Natural form:** position 1 is the service digit, which also shows the sign of the number. Positions 2 to 9 are the eight digit positions holding the number.

Table 1. Permitted argument range and accuracy of functions

Function	Permitted argument range	Max. absolute error	Max. relative error, %
sin x	0 ⩽ x ⩽ 90°; 0 < x < 1.571 rad	±1·10⁻⁴	1·10⁻³
cos x	0 ⩽ x ⩽ 90°; 0 < x < 1.571 rad	±1·10⁻⁴	1·10⁻³
tg x	0 ⩽ x < 90°; 0 < x < 1.571 rad	±1·10⁻⁴	1·10⁻³

Function	Permitted argument range	Max. absolute error	Max. relative error, %
arcsin x	0 ⩽ x ⩽ 1	±5·10⁻⁴	1·10⁻²
arccos x	0 ⩽ x ⩽ 1	±5·10⁻⁴	1·10⁻²
arctg x	0 ⩽ x ⩽ 500	±5·10⁻⁴	1·10⁻²
y^x	Result must lie between ±1·10⁻⁷ and ± (1·10⁸ − 1)	−20124 [as printed]	5·10⁻²
e^x	|x| < 230.2585	6 [as printed]	1·10² [as printed]
x²	1·10⁻⁹⁹ < x < 99999·10⁴⁵	0	0
10^x	−99.999999 ⩽ x ⩽ 99.999999	0	0
1/x	1·10⁻⁹⁹ ⩽ x ⩽ 9.9999·10⁹⁹	±1·10⁻⁸	5·10⁻³
√x	1·10⁻⁹⁹ ⩽ x ⩽ 9.9999·10⁹⁹	0	0
ln x	1·10⁻⁹⁹ ⩽ x ⩽ 9.9999·10⁹⁹	±1·10⁻⁴	1·10⁻³
lg x	1·10⁻⁹⁹ ⩽ x ⩽ 9.9999·10⁹⁹	±1·10⁻⁴	1·10⁻³

3. Main technical characteristics

Item	Characteristic	Value
3.1	Number of digits, fixed or natural decimal point	8
	Number of digits, floating point	Mantissa 5, exponent 2
3.2	Number range, natural decimal point	±10⁻⁷ ⩽ x ⩽ ±(10⁸ − 1)
	Number range, floating point	±10⁻⁹⁹ ⩽ x ⩽ ±9.9999·10⁹⁹
3.3	Number of operations performed automatically	29
3.4	Average time for an arithmetic operation	0.35 s
	Average time to extract a square root	0.4 s
3.5	Number of memory registers	1
3.6	Operating temperature	10 °C to 35 °C

Item	Characteristic	Value
	Relative humidity	40 to 80% at 25 °C
	Atmospheric pressure	84.0 to 107 kPa
3.7	Power consumption	not more than 9.0 W
3.8	Overall dimensions, not more than	Length 250 mm, width 180 mm, height 78 mm
3.9	Mass, not more than	1.0 kg
3.10	Power supply	42 V AC +10% / −15%, at (50 ± 1) Hz

4. Contents of delivery

- «Электроника МКШ 2» microcalculator: 1
- Operating manual: 1 copy
- Individual consumer packaging: 1 set

5. Safety instructions

5.1. The live parts of the microcalculator must be reliably insulated.

5.2. The microcalculator may be inspected and repaired only by persons who have received safety instruction and have studied its design and operation.

5.3. At the end of calculations, and if any fault appears, switch the microcalculator off and disconnect the plug of its lead from the supply.

6. Design of the microcalculator

6.1. The «Электроника МКШ 2» microcalculator is a sequential-action computing machine.

6.2. The external appearance of the microcalculator is shown in Figure 2. The case is made of high-impact polystyrene. [Figure 2 shows the desk-top case, with the display window at the top, the keyboard below it and the power switch on the right-hand side.]

6.3. Numbers are entered and operations performed on them by pressing the appropriate keys. The keys have two functions each. The key layout is shown in Figure 3 and their purpose is given in Table 2.

6.4. A check test of the microcalculator's operation is given in Table 3.

6.5. The circuit diagram of the microcalculator is given in Appendix 1.

Figure 3. Keyboard layout

Each cell shows the key's main legend, with its second function (selected with **F**) in brackets.

7 (sin)	8 (cos)	9 (tg)	÷ (1/x)	x→П (Δ)	Cx (CF)
4 ($\sin^{-1}$)	5 ($\cos^{-1}$)	6 (tg^{-1})	× (x^2)	П→x ($x_{1,2}$)	F
1 (ln)	2 (lg)	3 (π)	− (x↔П)	ВП (y^x)	(
0 (e^x)	, (10^x)	/−/ (CN)	+ (П+)	= (V)	)

Table 2. Key legends and functions

Key	Direct function	Function after pressing F
F	Selects the second function (shift)	—
(	Opening bracket. Also enters the "unknown" marker when solving a quadratic equation or a system of two linear equations in two unknowns	—
)	Closing bracket	—
/−/ (CN)	Changes the sign of the number or of the exponent	Changes the form in which the number is displayed
7 (sin)	Enters the digit 7	Sine
8 (cos)	Enters the digit 8	Cosine
9 (tg)	Enters the digit 9	Tangent
÷ (1/x)	Division	Reciprocal
П→x ($x_{1,2}$)	Recalls the number from memory register П	Roots of a quadratic equation
4 ($\sin^{-1}$)	Enters the digit 4	Arcsine
5 ($\cos^{-1}$)	Enters the digit 5	Arccosine
6 (tg^{-1})	Enters the digit 6	Arctangent
× (x^2)	Multiplication	Square of the number

Key	Direct function	Function after pressing F
x→П (Δ)	Stores the number in memory register П	Determinant of a system of equations
1 (ln)	Enters the digit 1	Natural logarithm
2 (lg)	Enters the digit 2	Common (base-10) logarithm
3 (π)	Enters the digit 3	Recalls the constant π
− (x↔П)	Subtraction	Exchanges the contents of registers X and П
ВП (y^x)	Enters the exponent of a number	Raises y to the power x
0 (e^x)	Enters the digit 0	Exponential function e^x
, (10^x)	Enters the decimal point	Power of ten, 10^x
= (√)	Gives the result	Square root
+ (П+)	Addition	Adds to memory register П
Cx (CF)	Clears register X (the display)	Cancels the second-function (F) flag

Table 3. Functional check test

Press the keys in order and compare the display after each one. The Г–Р column shows where to set the degrees (Г) / radians (Р) switch before that step. In the Display column, a space separates the mantissa from the exponent, which occupies positions 6 to 8.

No.	Г–Р	Key	Sign	Display, positions 1–8
1		Cx		0.
2		1		1.
3		2		12.
4		0		120.
5		÷		120.
6		4		4.
7		=		30.
8	Г	F		30.

No.	Г–Р	Key	Sign	Display, positions 1–8
9		tg		5.7735 −01
10		F		5.7735 −01
11		tg^{-1}		30.
12		÷		30.
13		1		1.
14		5		15.
15		,		15.
16		7		15.7
17		0		15.70
18		8		15.708
19		=		1.9098549
20		F		1.9098549
21		1/x		5.2359 −01
22		F		5.2359 −01
23		CN		52359998 [printed without a point]
24	Р	F		52359998
25		tg		5.7735 −01
26		×		5.7735 −01
27		1		1.
28		,		1.
29		5		1.5
30		=		8.6602 −01
31	Г	F		8.6602 −01
32		cos^{-1}		30.
33		F		30.
34		П+		30.
35		ВП		30. 00

No.	Г–Р	Key	Sign	Display, positions 1–8
36		1		30. 01
37		/–/		30. –01
38		F		30. –01
39		10^x		1000.
40		F		1000.
41		П+		1000.
42		F		1000.
43		lg		3.
44		П→x		1030.
45		F		1030.
46		√		32.093613
47		F		32.093613
48		ln		3.4687
49		ВП		3.4687 00
50		9		3.4687 09
51		9		3.4687 99
52		x→П		3.4687 99
53		×		3.4687 99
54		6		6.
55		=	–	– – – – – – – – (overflow)
56		6		6.
57		–		6.
58		(		6.
59		8		8.
60		–		8.
61		4		4.
62		)		4.

No.	Г–Р	Key	Sign	Display, positions 1–8
63		=		2.
64		(		2.
65		3		3.
66		(		3.
67		1		1.
68		4		14.
69		/–/	–	14.
70		F	–	14.
71		$x_{1,2}$	–	3.5
72		F	–	3.5
73		$x_{1,2}$		2.
74		П→x		3.4687 99

Note: repeat the check test only after all registers have been cleared. To do this, switch the microcalculator off and on again.

7. Preparation for use

7.1. To fix the microcalculator to a desk that has a 42 V supply (a socket fitted), drill three recesses in the desk as shown in Figure 4. Drill them first with a fine drill, then with a 15 mm drill to a depth of 11 mm. Insert the pins fitted to the calculator's bottom cover into the recesses. The pins stop the microcalculator from sliding on the desk.

Figure 4. Marking out for fixing the microcalculator to the desk

- Three holes, ø 15.0 mm, depth at least 11 mm.
- Two holes on one line, **120 ± 0.4 mm** apart, centred on the calculator's centre line.
- The third hole on the centre line, **170 ± 0.5 mm** from the line through the other two.

7.2. To connect the microcalculator to the 42 V AC supply, the recommended equipment is the mathematics-classroom electrical kit type КЭМ, or other similar kits that distribute 42 V AC to the pupils' desks and the teacher's desk.

7.3. Connect the microcalculator to the 42 V AC supply with its plug. Switch the microcalculator on with the switch on the right-hand side of the case.

The calculator is ready for use when the display shows **a zero and a decimal point in the most significant position**. If anything else is displayed, press **Cx**, or switch the microcalculator off and on again after waiting at least 3 seconds.

8. Operating procedure

8.1. Supervision

8.1.1. Pupils must carry out all calculations on the microcalculator under a teacher's supervision.

8.1.2. The microcalculator may be left switched on for as long as the lesson requires. After switching off, wait at least 3 seconds before switching on again.

8.2. Entering numbers

8.2.1. Enter numbers in natural form by pressing the digit keys and the **,** (decimal point) key in the order they are written. Entry is locked once eight digits have been entered, so further digit keys are ignored.

Attention! For a negative number, enter its absolute value and then press **/−/**.

8.2.2. For a number in normal (exponential) form, first enter the mantissa and its sign as in 8.2.1, then press **ВП** and enter the exponent. If the exponent is negative, press **/−/**. You can press **/−/** either before or after entering the exponent.

Note: if a mantissa with more than five digits is entered, the display shows only its five most significant digits, and two zeros appear in the 8th and 9th positions reserved for the exponent. The calculator still holds the full mantissa as entered.

8.2.3. If a number is shown as a mantissa and exponent during entry or calculation, pressing **F** then **CN** shows the eight-digit mantissa. Pressing them again returns to a mantissa with the point after the first digit, plus the exponent.

8.2.4. If you make a mistake entering a number, press **Cx** and enter it again. If the mistake was in the exponent, simply enter it again. Each new digit goes into the lower digit of the exponent, the previous digit moves one place left, and the upper digit is lost. The mantissa is unchanged.

8.2.5. Check the number entered, and the result, on the display.

8.3. Simple arithmetic

1. Enter the first number.
2. Press the key for the operation.

3. Enter the second number.
4. Press **=**.
5. Read the result on the display (see Table 4).

Table 4

Example	Key sequence	Result
2 + 3	2 + 3 =	5
−3.5 × 6	3 , 5 /−/ × 6 =	−21
10 − 6	10 − 6 =	4
40 ÷ 8	40 ÷ 8 =	5

8.4. Chained arithmetic

8.4.1. In chained calculations, the result of one operation is the first number of the next.

8.4.2. Enter numbers and operations in the order they are written, taking account of operator precedence (see Table 5).

Table 5

Example	Key sequence	Result
10 ÷ 2 + 3	1 0 ÷ 2 + 3 =	8
3 × 4 − 6	3 × 4 − 6 =	6

8.5. Calculations with a constant

8.5.1. The number entered after an operation key is stored as a constant.

8.5.2. To calculate with a constant:

1. Enter the first number.
2. Press the operation key.
3. Enter the constant.
4. Press **=**.
5. Read the result.
6. Enter the next number.
7. Press **=**.

8. Read the result (see Table 6).

Table 6

Example	Key sequence	Result
4 × 3	4 × 3 =	12
7 × 3	7 =	21
12 × 3	1 2 =	36

8.5.3. The constant can be used to raise a number to a whole-number power:

1. Enter the base.
2. Press **×**.
3. Enter the base again.
4. Press **=** n − 1 times, where n is the power.
5. Read the result (see Table 7).

Table 7

Example	Key sequence	Result
3^4	3 × 3 = = =	81
5^3	5 × 5 = =	125

8.6. Recalling the constant π

1. Press **F**.
2. Press **π** (the 3 key).
3. Read the value of π on the display.

8.7. Functions x^2, 1/x, √x, e^x, 10^x, ln x, lg x, sin x, cos x, tg x, arcsin x, arccos x, arctg x

1. Enter the value of x.
2. Press **F**.
3. Press the key marked with the required function.
4. Read the result on the display (see Table 8).

Table 8

Example	Key sequence	Result	Note
5^2	5 F x²	25	
1/4	4 F 1/x	$2.5{\cdot}10^{-1}$	
√121	1 2 1 F √	11	
e^1	1 F e^x	2.7183	
10^3	3 F 10^x	1000	
ln 5	5 F ln	1.6094	
lg 1000	1 0 0 0 F lg	3	
sin 30°	3 0 F sin	$5{\cdot}10^{-1}$	Г–Р switch at Г (degrees)
cos 60°	6 0 F cos	$5{\cdot}10^{-1}$	Same
tg 45°	4 5 F tg	1	Same
arcsin 0.5	0 , 5 F $\sin^{-1}$	30°	Same
arccos 0.5	0 , 5 F $\cos^{-1}$	60°	Same
arctg 1	1 F tg^{-1}	45°	Same
sin 4.71239 rad	4 , 7 1 2 3 9 F sin	−1 [printed as 1]	Г–Р switch at Р (radians)
cos 1 rad	1 F cos	$5.403{\cdot}10^{-1}$	Same

8.8. The function y^x

1. Enter the base y.
2. Press **F**.
3. Press **y^x** (the ВП key). The display shows the natural logarithm of the base.
4. Enter the exponent x.
5. Press **=**.
6. Read the result (see Table 9).

Table 9

Example	Key sequence	Result
5^3	5 F y^x 3 =	124.99
3^4	3 F y^x 4 =	80.996

Attention! The **y^x** key computes y^x from a power series, so the result carries an error (see Table 1). To raise a number to a whole-number power with no error, use the constant method in 8.5.3.

8.9. Calculations with brackets

8.9.1. In arithmetic with brackets, no more than three brackets may be open in succession.

8.9.2. y^x must be calculated before the first opening bracket, or after an arithmetic operation sign that follows the third closing bracket.

8.9.3. e^x, 10^x, ln x and lg x may be calculated only when the number of open brackets exceeds the number of closed ones by no more than 1.

8.9.4. Trigonometric and inverse trigonometric functions, and recalling π, may be used only when the number of open brackets exceeds the number of closed ones by no more than 2.

8.9.5. For an example of a calculation with brackets, see Table 10.

Table 10

Example	Key sequence	Result
$3^4 \times (\sqrt{121} \div (\sin 30° + 1/4) - \lg 57)$	3 F y^x 4 × (1 2 1 F √ ÷ (3 0 F sin + 4 F 1/x) − 5 7 F lg) =	1045.7204

8.10. Solving a quadratic equation

8.10.1. The calculator finds the real roots of a quadratic equation in canonical form, $ax^2 + bx + c = 0$.

8.10.2. To find the roots:

1. Enter coefficient a and its sign.
2. Press **(**.
3. Enter coefficient b and its sign.
4. Press **(**.
5. Enter the constant term c and its sign.
6. Press **F**.

7. Press **$x_{1,2}$** (the П→x key).
8. Read the root x_1 on the display.
9. For the root x_2, repeat steps 6, 7 and 8 (see Table 11).

Table 11

Example	Key sequence	Result
$5x^2 - 7x - 4 = 0$	5 (7 /−/ (4 /−/	
	F $x_{1,2}$	$x_1 = -4.3578 \cdot 10^{-1}$
	F $x_{1,2}$	$x_2 = 1.8357816$

8.10.3. If the equation has no real roots, the display shows the overflow sign.

8.11. Solving a system of two linear equations in two unknowns

8.11.1. Solving the system $a_1x + b_1y = c_1$, $a_2x + b_2y = c_2$ comes down to calculating its determinants:

```
\Delta = \begin{vmatrix} a_1 & b_1 \\ a_2 & b_2 \end{vmatrix} \qquad \Delta_1
= \begin{vmatrix} c_1 & b_1 \\ c_2 & b_2 \end{vmatrix} \qquad \Delta_2 =
\begin{vmatrix} a_1 & c_1 \\ a_2 & c_2 \end{vmatrix} \qquad x =
\frac{\Delta_1}{\Delta} \qquad y = \frac{\Delta_2}{\Delta}
```

8.11.2. To find the determinant Δ:

1. Enter coefficient a_1 and its sign.
2. Press **(**.
3. Enter coefficient b_1 and its sign.
4. Press **(**.
5. Enter coefficient a_2 and its sign.
6. Press **(**.
7. Enter coefficient b_2 and its sign.
8. Press **F**.
9. Press **Δ** (the x→П key).
10. Store Δ in the memory register by pressing **x→П**.

8.11.3. To find Δ_1 and x:

1. Enter c_1 and its sign.
2. Carry out steps 2, 3 and 4 of 8.11.2.

3. Enter c_2 and its sign.
4. Carry out steps 6, 7, 8 and 9 of 8.11.2.
5. Press **÷**.
6. Recall Δ from memory by pressing **П→x**.
7. Press **=**.
8. Read x on the display.

8.11.4. To find Δ_2 and y:

1. Carry out steps 1 and 2 of 8.11.2.
2. Enter c_1 and its sign.
3. Carry out steps 4, 5 and 6 of 8.11.2.
4. Enter c_2 and its sign.
5. Press **F**.
6. Press **Δ**.
7. Carry out steps 5, 6 and 7 of 8.11.3.
8. Read y on the display (see Table 12).

Table 12

Example	Key sequence	Result
4x + 5y = 14; 10x + 3y = 16	4 (5 (1 0 (3 F Δ x→П	
	1 4 (5 (1 6 (3 F Δ ÷ П→x =	x = 1
	4 (1 4 (1 0 (1 6 F Δ ÷ П→x =	y = 2

8.11.5. If the system has no solution, the display shows the overflow sign.

8.12. Memory operations

8.12.1. The microcalculator can store a number in the memory register, recall it, add to the memory register, and exchange the contents of the memory and display registers.

8.12.2. To store a number in memory, press **x→П**.

8.12.3. To recall it, press **П→x**.

8.12.4. To add to memory, press **F** then **П+**.

8.12.5. To exchange the memory and display registers, press **F** then **x↔П**.

8.12.6. For examples, see Table 13.

Table 13

Example	Key sequence	Result
7 × 3 + 12 ÷ 3	7 × 3 = x→П 1 2 ÷ 3 = F П+ П→x	25
3 × sin 30°	3 0 F sin x→П × 3 =	1.5
7 + sin 30°	7 + П→x =	7.5
4 ÷ sin 30°	4 ÷ П→x =	8
(9 + 5) ÷ (9 − 5)	9 x→П − 5 = F x↔П + 5 = ÷ П→x =	3.5

8.13. Overflow

8.13.1. Overflow occurs:

1. when a result exceeds $9.9999 \cdot 10^{99}$ in magnitude;
2. on an invalid operation, such as division by 0, the square root of a negative number, the logarithm of a negative number, or ln 0;
3. when numbers outside the working range are entered (see Table 1).

8.13.2. In overflow, the display shows «—» in all positions.

8.13.3. To continue calculating, clear the overflow by pressing **Cx**.

8.13.4. If a result is smaller than $1 \cdot 10^{-99}$ in magnitude, the display shows zero.

8.13.5. If **Cx** is pressed during a calculation, after an operation key, the microcalculator stops performing subtraction, multiplication and division. To restore normal operation, switch it off and on again and reset it with **Cx**, or, without switching off, press **+**.

9. Storage rules

9.1. Store microcalculators in dry, heated rooms, free of corrosive substances in the air and out of direct sunlight. Relative humidity must not exceed 90% at temperatures from +1 °C to +40 °C.

9.2. In winter, leave the microcalculator at room temperature for 2 hours before use.

10. Warranty

10.1. The manufacturer guarantees that the microcalculator meets technical specification ПМО.081.250 ТУ, provided the user follows the operating rules in this manual.

10.2. The warranty period is 24 months from the date the microcalculator is put into service.

10.3. The service life of the microcalculator is at least 10 years.

10.4. If the microcalculator fails during the warranty period, the owner is entitled to free repair on presentation of the warranty coupon.

10.5. Without the warranty coupon, or if the seals on the microcalculator have been broken, no quality claims are accepted and no warranty repair is carried out.

10.6. During the warranty period, repairs are at the owner's expense if the microcalculator has not been operated in accordance with this manual, or if the repair organisation's recommendations for keeping it in good working order have not been followed.

Appendix 1. Circuit diagrams

The original has two schematic sheets, which are reproduced from the scan at the end of this appendix. Below are the translated labels, notes and pinout tables. The scan is poor in places, and readings that are uncertain are marked [?].

Sheet 1: microcalculator circuit diagram (manual page 29)

The calculator has three assemblies connected by plated-through holes X1–X3:

Assembly	Name	Main parts
E1	Main (операционное) board	IV-18 (ИВ-18) display; D1 К165ГФ3 clock generator; D2 К145ИК1301 processor; D3–D5 К161КН1Б display drivers; pull-down resistors R1–R35; R38 and VD2 (Note 8)
E2	Input (keyboard) board	S1 key switches; S2 Г–Р switch [?]; diodes VD4, VD5
E3	Power supply Э2-21	See sheet 2

Connector X2 (power supply to main board)

X2 pin	Circuit	Label
1	Б	−15 V
2	А	⊥ common
3	Г	−45 V
4	Д	Filament (Накал)

X2 pin	Circuit	Label
5	Е	Filament
6	—	Ср.Т (filament winding centre tap)

Notes on sheet 1

1. Resistors R1–R35, R37 and R38 are type МЛТ-0.125. Capacitor C1 is К10-7В-Н70 and C2 is К10-7В-Н90.
2. * Selected during adjustment.
3. X1, X2 and X3 are plated-through holes in the board.
4. Connect pin 1 of D1 and D3–D5 to circuit А (⊥, common).
5. Connect pin 7 of D1 and pin 48 of D2 to circuit Б (−15 V).
6. Connect pin 9 of D3–D5 to circuit Г (−45 V).
7. Diode Д9Г may be replaced by Д9Б, Д9В or Д9Д.
8. ** Fitted during adjustment at the factory if unused digits of the display glow. [This refers to R38, 27 kΩ, and VD2, Д814А.]

Sheet 2: continuation of Appendix 1 (manual page 30)

Power supply unit Э2-21

The 42 V AC input passes through the on/off switch (marked Вкл / Выкл) and fuse FU to winding I (pins 1–2) of transformer T. Winding II (pins 4–5) feeds bridge rectifier VD1–VD4 (КД105В), which gives the −45 V rail. A series regulator (VT1 МП26Б, VT2 КТ801А, R1 27 kΩ, zeners VD5 КС147А, VD6 КС147А and VD7 КС168А) gives the −15 V rail. Winding III (pins 6, 3 and 7) supplies the display filament.

Output pin	Circuit
1	−15 V
2	+ common
3	−45 V
4	~5 V
5	~5 V
6	~24 [as printed; this is the centre tap, Ср.Т, on sheet 1]

Notes on the power supply:

1. Zener VD7 may be replaced by KC147A.
2. Zeners VD5 and VD6 may be replaced by two Д814А, or by Д814Б and КС168А, or by Д814В and КС156А.
3. In supply units with a −27 V output, zener VD8 type КС168А is added in series with VD7, and VD5 and VD6 are type КС156А.
4. Capacitors C1–C4 are К50-6-II-100 V.

К165ГФ3 pinout (4-phase pulse generator)

Pin	Function
1	Case
2	Stabiliser input
3	Generator output Ф1
4	Generator output Ф2
7	Supply, −15 V
9	Generator output Ф4
10	Generator output Ф3
11	Generator adjustment input

Pins not listed are unused. The pulse repetition frequency is set by resistor R37.

К145ИК1301 pin functions (48-pin package)

Pins	Signals	Function
1, 2, 3, 4	Ф4, Ф1, Ф3, Ф2	Pulse supply inputs from the generator
8, 9, 10, 15, 16, 17, 18, 20, 21, 22, 23	Д12, Д11, Д10, Д9, Д8, Д7, Д6, Д5, Д4, Д3, Д2	Timing signal outputs and display digit control
24	—	Case
25, 26, 28	[?]	Inputs
29	[?]	Output
30, 31	К2, К1	Keyboard inputs
32, 33	[?]	Mode inputs and outputs

Pins	Signals	Function
39, 41–47	Segment outputs [?]	Display anode (segment) control outputs
48	—	Supply

Pins not listed are unused.

Microcircuit connections and operating mode

Ref	Type	Function	Common (⊥)	−15 V
D1	К165ГФ3	Pulse generator	Pin 1	Pin 7
D2	К145ИК1301	Control and processing	Pin 24	Pin 48

К161КН1Б pinout: 1 common; 2–8 outputs 1–7; 9 supply voltage; 10–16 inputs 7 to 1.

IV-18 (ИВ-18) display: F and F_e are the cathode (filament) leads; G1–G9 are the grids of digit positions 1–9. A table in the original lists each display lead and the circuit that drives it.

Segment pins used to form digits and signs

Display pins	Character
3, 4, 5, 10, 11, 12	0
4, 10	1
3, 5, 9, 11, 12 [one value obscured]	2
3, 4, 9, 10, 12	3
4, 9, 10, 11	4
3, 4, 9, 11, 12	5
3, 4, 5, 9, 11, 12	6
4, 10, 12	7
3, 4, 5, 9, 10, 11, 12	8
3, 4, 9, 10, 11, 12	9
2	Decimal point («.»)
9	Minus sign

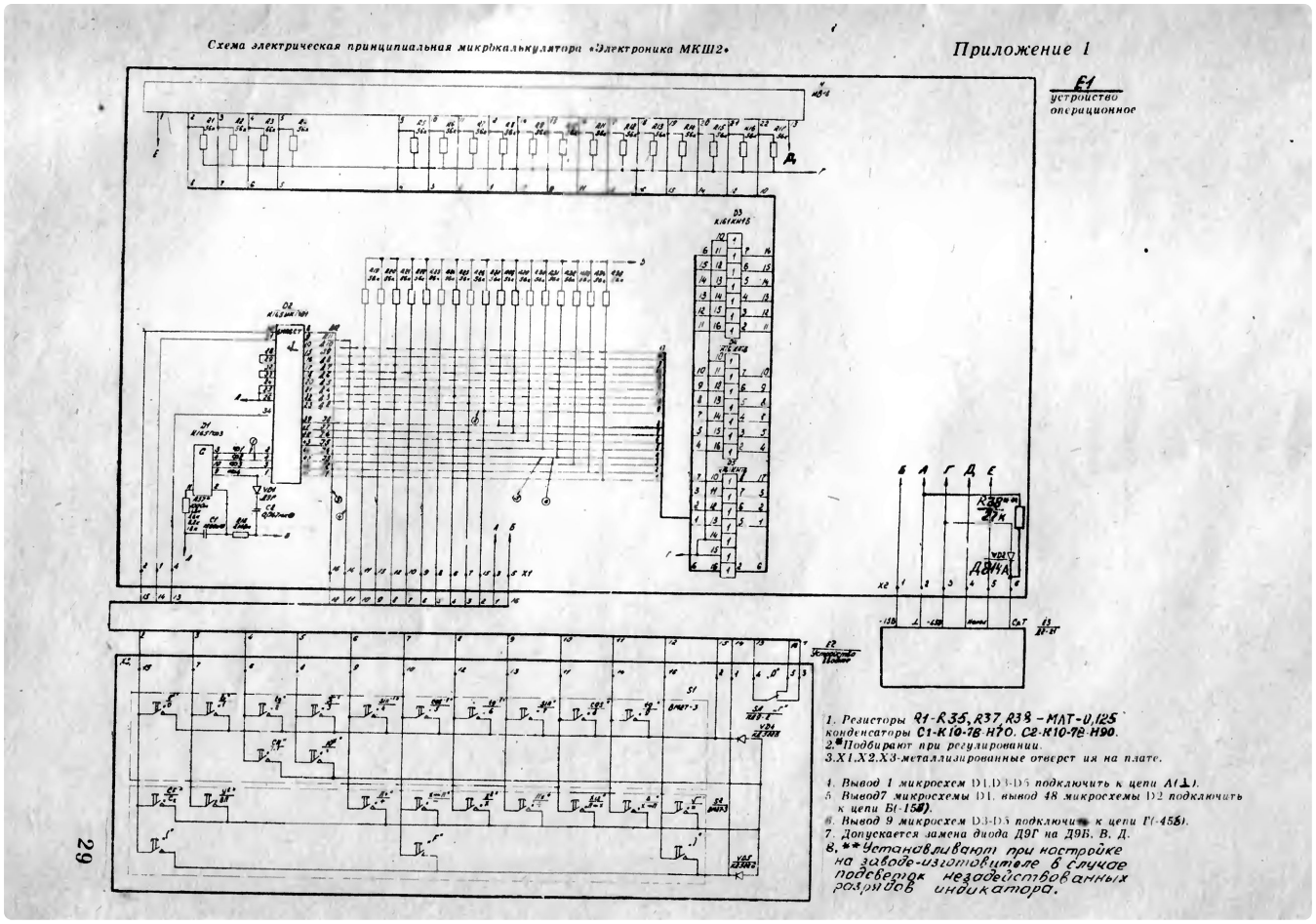
Oscilloscope traces: the original shows six traces and a timing diagram of digit signals Д2 to Д12. Marked timings include 2 µs, 10 µs and 3.5 µs on trace 1, and 0.16 ms and 2.44 ms on traces 2 and 5. The notes read:

- Trace 3: signal when key **9** is pressed.
- Trace 4: signal when key **−** is pressed [?].
- Trace 5: Д4 with “0” on the display.
- Trace 6: Д4 with all “8” on the display.

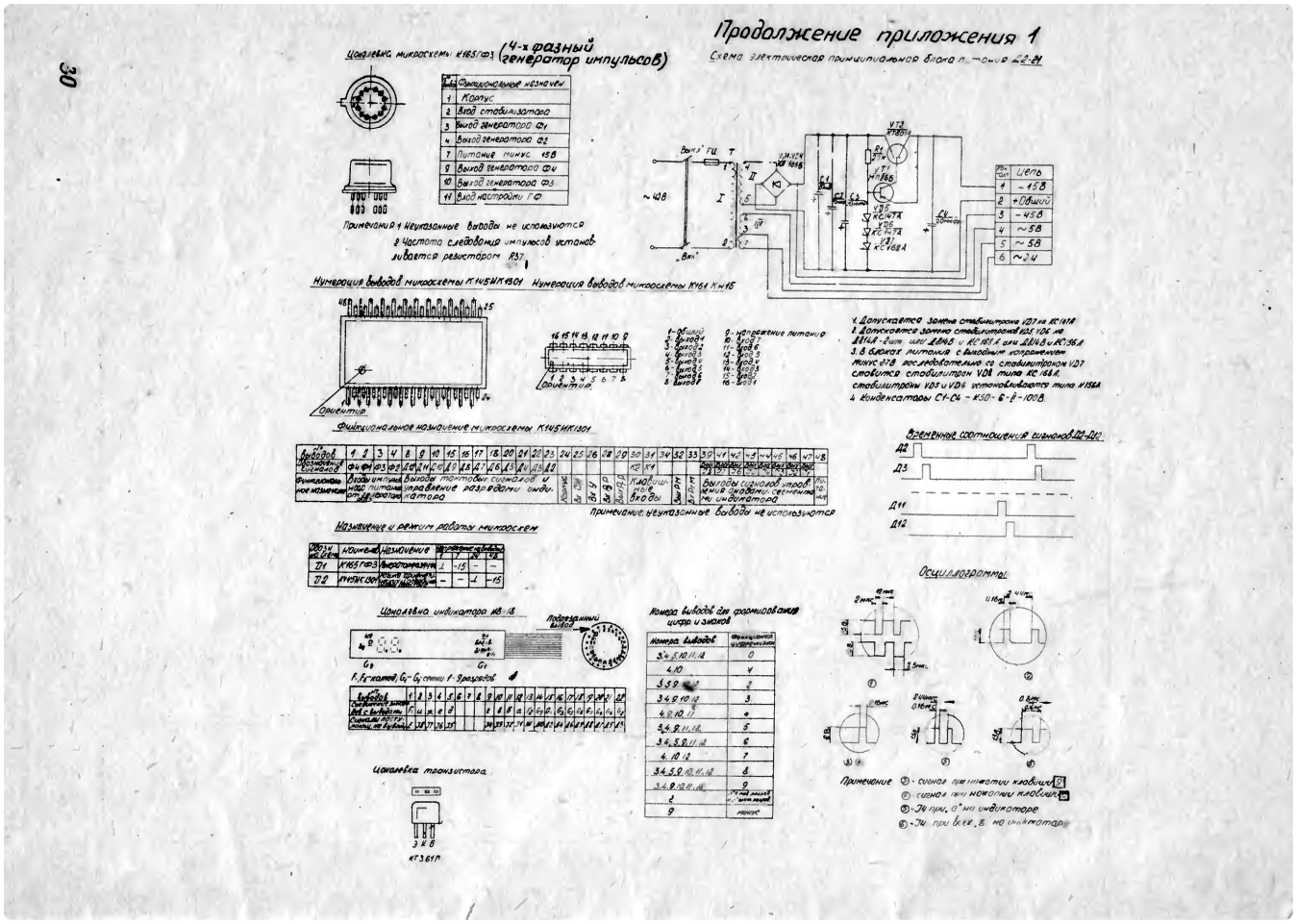
Transistor КТ361Г: the pinout drawing shows the leads as E, C, B (Э, К, Б).

Scans of the original sheets

Sheet 1, manual page 29: microcalculator circuit diagram.



Sheet 2, manual page 30: power supply Э2-21, IC pinouts, display pinout and oscilloscope traces.



Warranty repair coupons (manual pages 27–28)

The manual contains two warranty repair coupons, Coupon 1 and Coupon 2. Each has a half that stays with the owner (остается у потребителя) and a half that is torn off by the repairer (отрывается). In this copy they are stamped by the factory quality-control department (OTK), with a release date that appears to read October 1988 [?]. The fields are:

1. Microcalculator «Электроника МКШ 2», No. ____
2. Date of manufacture ____ 19__
3. Factory quality-control (OTK) representative: signature, stamp
4. Date put into service ____ 19__
5. Address and name of the user organisation
6. Is the microcalculator under technical maintenance? (yes, no, by whom)
7. Operating hours before failure
8. External symptoms of the failure
9. Main causes of the failure

The reverse of each coupon (“continuation of coupon”) has:

10. Parts, assemblies and materials used
11. Microcalculator returned to service after repair: when, by whom

It also has spaces for the signatures of the maintenance specialist and the head of the organisation, a place for the official stamp (М.П.) and the date.

Appendix 2. Feedback form: performance of the «Электроника МКШ 2» microcalculator

- Microcalculator «Электроника МКШ 2» No. ____, date of manufacture ____
- Where the product is installed (school, vocational school (ПТУ), higher-education institution (ВУЗ), etc.)
- Period of use: from ____ to ____
- Approximate hours per day the microcalculator was used
- Does the microcalculator work reliably? Does it meet all its technical and quality specifications?
- Have there been any failures or faults?
- Describe the external symptoms of the failures and faults.
- Approximate operating hours before a failure occurred
- Approximate operating hours during the warranty period
- Your comments and suggestions on the product's performance, ease of use and appearance

Please send the feedback form to: 317000, Svetlovodsk, Kirovograd Oblast, P.O. Box No. 7.

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The original was printed by the Globino district printing house (Глобинская районная типография).